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ORIGINAL ARTICLE

Social Media Use and Adolescent Depressive Symptoms: The Moderating Roles of Gender and Parental Monitoring

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Abstract
Social media use has significantly reshaped adolescents' social experiences, raising substantial concerns about its impact on mental health. This study examines the association between social media usage intensity and depressive symptoms among adolescents, emphasizing the moderating roles of parental monitoring and gender. Using logistic regression analysis of the 2023 US Youth Risk Behavior Survey ($N = 10{,}334$), results reveal that moderate and high social media use significantly increases depressive symptoms. Notably, higher parental monitoring mitigates these adverse effects. While females exhibit higher depressive symptoms overall, gender does not significantly moderate social media's impact. This study highlights the importance of nuanced parental engagement and offers critical insights for policymakers and educators to foster healthier digital environments for adolescents.

Keywords Social media usage · Parental monitoring · Adolescent mental health · Depressive symptoms · Gender difference

The rapid proliferation of social media platforms has significantly altered how adolescents interact, socialize, and perceive themselves (Boyd, 2014). While social media can facilitate communication and foster a sense of community, mounting evidence suggests that its excessive use is linked to adverse mental health outcomes, particularly depressive symptoms in teenagers (Murthy, 2023; Shannon et al., 2022). According to recent studies, adolescents who spend more than 3 h per day on social media are at a heightened risk of developing mental health issues, such as depression and anxiety, compared to their peers who engage less frequently with these platforms (Primack et al., 2017, 2021; Twenge & Campbell, 2018; Twenge et al., 2018).

One of the primary concerns is the role of social comparison, where adolescents often measure their self-worth against the idealized representations of others' lives on platforms

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like Instagram and TikTok. This constant exposure can exacerbate feelings of inadequacy, loneliness, and social isolation, contributing to the onset of depressive symptoms (Valkenburg et al., 2022; Yigiter et al., 2024). Furthermore, social media features such as likes, comments, and followers create a feedback loop of social validation that can amplify feelings of low self-esteem and trigger depressive episodes, especially in impressionable teenage users (Primack et al., 2021; Sherlock & Wagstaff, 2019).

Adolescence is a formative period marked by heightened sensitivity to social evaluation and identity development (Blakemore & Mills, 2014; Sommerville, 2013). This neurodevelopmental stage is characterized by a heightened reactivity in the brain's socio-affective circuits, particularly the ventral striatum and medial prefrontal cortex, which makes adolescents exquisitely sensitive to peer acceptance and rejection (Blakemore & Mills, 2014; Meshi et al., 2015). The intrinsic search for validation is a core part of this process; it is a fundamental psychological need to feel accepted, worthy, and competent within one's social group (Baumeister & Leary, 1995). Social media platforms effectively hijack this intrinsic need by providing a quantified, public, and constant metric of social validation (e.g., likes, shares, follower counts), transforming abstract social approval into tangible reinforcement (Crone & Konijn, 2018). For adolescents, this online validation can become a primary source of self-worth, making them vulnerable to a cycle where their mood and self-esteem become contingent on external, often unpredictable, online feedback (Nesi & Prinstein, 2015).

Social media platforms intensify these processes by providing environments where adolescents can engage in constant self-presentation and peer comparison. While social media can foster connections and provide support networks, it also facilitates upward social comparison, where users measure their lives against the curated, idealized portrayals of others. This can lead to feelings of inadequacy, lower self-esteem, and increased vulnerability to depressive symptoms (Huang, 2017; Verduyn et al., 2020). The interplay between the quest for validation and the experience of social comparison creates a potent psychological environment where mental well-being becomes heavily contingent on online feedback.

Moreover, the rapid proliferation of social media use has outpaced our understanding of its psychological effects. Recent longitudinal and experimental research has begun to clarify the direction of this relationship, suggesting that social media use is a significant predictor of future depressive symptoms, rather than merely a symptom of existing distress (Primack et al., 2021; Davis & Goldfield, 2025). By exploring the connections between social media usage and mental health, this study aims to fill critical gaps in knowledge, enabling informed decisions and interventions to support adolescent well-being.

Theoretical Considerations

The relationship between social media use and mental health, particularly depressive symptoms among adolescents, is a growing area of research (Twenge et al., 2018). This expanding body of work highlights the intricate ways digital platforms influence emotional well-being, with both positive and negative consequences. Understanding this relationship requires a thorough examination of psychological theories, social behaviors, and the digital landscape that shapes adolescent interactions.

Originally developed by Festinger (1954), social comparison theory posits that individuals have an innate drive to evaluate their own abilities, opinions, and self-worth by comparing themselves to others. In the context of social media, this theory becomes particularly

relevant. Adolescents are highly susceptible to comparing themselves to the often idealized and curated representations of peers and influencers on platforms like Instagram and TikTok. These comparisons can lead to negative self-evaluations and feelings of inadequacy, ultimately contributing to depressive symptoms (Vogel et al., 2014). A study by Perloff (2014) found that social media exacerbates self-comparison due to the constant exposure to others' highlight reels, which are often filtered and selectively presented. Adolescents, whose self-identity is still developing, are particularly vulnerable to these comparisons. The study highlighted a strong correlation between frequent social media use and lower self-esteem, body dissatisfaction, and symptoms of depression, especially among teenage girls. Furthermore, Appel et al. (2016) noted that upward social comparisons (i.e., comparing oneself to someone perceived as better off) on social media are significantly associated with negative emotional outcomes. Adolescents who constantly encounter posts about others' successes, vacations, and seemingly perfect lives may develop a distorted perception of reality, leading to depressive thoughts and feelings of loneliness.

Direct evidence from a study by Nesi and Prinstein (2015) underlines the role of social media in shaping adolescents' self-concept. The research shows that not only do adolescents compare themselves to peers, but they also experience anxiety and depression stemming from perceived social pressure to match the curated online personas of others. Social media plays a crucial role in shaping adolescents' self-perception, which refers to how individuals view themselves. In the digital landscape, adolescents actively construct and curate their identities, often seeking validation through metrics such as likes, comments, and follower counts. This behavior reflects their developmental stage, which emphasizes peer acceptance and social comparison (Valkenburg & Peter, 2013). While receiving positive feedback can momentarily enhance self-esteem, negative feedback or a perceived lack of engagement can lead to self-doubt and a negative self-image, creating a fragile sense of self-worth (Nesi & Prinstein, 2015).

A pressing concern is the impact of social media on adolescents' body image, particularly on platforms like Instagram, where visual content dominates. Adolescents are frequently exposed to idealized and edited images that promote unattainable beauty standards. This exposure can heighten body dissatisfaction, especially among girls. Research by Jones and Taylor (2019) highlights that frequent engagement with appearance-centric content is strongly associated with increased body dissatisfaction, which is a known risk factor for depressive symptoms. Although positive reinforcement from social media can temporarily boost self-esteem, this gratification is often fleeting. Adolescents who rely heavily on external validation may enter a cycle of dependency, wherein their well-being becomes increasingly tied to others' feedback. This can exacerbate feelings of insecurity and vulnerability. Sherlock and Wagstaff (2019) explain that this external locus of control over self-worth makes adolescents particularly susceptible to emotional distress when their expectations are unmet.

The uses and gratifications theory, developed by Blumler and Katz (1974), focuses on why individuals actively seek out specific media to satisfy particular needs. This theory is useful for understanding adolescents' motivations for using social media and how these motivations can influence their mental health. According to this theory, teenagers may turn to social media for entertainment, social interaction, self-expression, or alleviating boredom (Joinson, 2008). These motivations are influenced by developmental and social factors unique to adolescence, such as the quest for identity and peer approval. Furthermore, social media platforms are designed to engage users through intermittent rewards, such as likes, comments, and notifications. These triggers activate the brain's dopamine system, creating a cycle of reward-seeking

behavior (Meshi et al., 2015). Over time, reliance on social media for social validation can contribute to emotional instability and increased susceptibility to mood disorders.

The gratifications obtained from social media use can vary widely, leading to both beneficial and detrimental psychological outcomes. On the positive side, social media can fulfill adolescents' need for connection and belonging. Platforms like Instagram and Snapchat offer opportunities for maintaining relationships, building social networks, and gaining social support, which can positively influence well-being (Best et al., 2014). Nesi et al. (2018) observed that adolescents often find a sense of community and emotional support through these interactions, which can buffer against stress and feelings of isolation.

On the contrary, excessive reliance on social media for gratification can have negative consequences. Adolescents may be exposed to harmful content, such as cyberbullying, social exclusion, and unrealistic beauty standards, which have been associated with increased depressive symptoms and anxiety (Keles et al., 2020; Nicholson et al., 2023; Young et al., 2024). Such experiences can undermine self-esteem and foster feelings of inadequacy.

The concept of fear of missing out (FoMO) is another crucial theoretical framework to consider. FoMO refers to the pervasive apprehension that others might be having rewarding experiences from which one is absent (Przybylski et al., 2013). This fear is intensified by the real-time nature of social media, where users are constantly bombarded with updates about social gatherings, parties, and other events they are not part of. Research by Baker et al. (2016) suggests that high levels of FoMO are linked to increased social media use, which, paradoxically, can lead to greater feelings of loneliness and depression. Adolescents may feel pressured to remain constantly connected to avoid missing out on social activities, leading to anxiety and disrupted sleep patterns, which are known risk factors for depression (Brown & Lee, 2022; Scott et al., 2019). The compulsive checking of social media feeds can create a cycle of addiction, where adolescents continuously seek updates, thereby reinforcing feelings of social exclusion and dissatisfaction. FoMO can lead to compulsive social media use, as adolescents continuously check their devices to stay updated and avoid feeling left out. This compulsive behavior can disrupt sleep patterns, academic performance, and overall well-being. Furthermore, sleep deprivation resulting from excessive social media engagement has been found to be negatively associated with mental health outcomes (Feng et al., 2024). Adolescents who spend excessive time on social media also often report reduced sleep quality and diminished real-world social connections, both of which are strong predictors of depressive symptoms (Orben et al., 2019).

These three theoretical perspectives—social comparison theory, uses and gratifications theory, and fear of missing out—converge on a common mechanism: social media creates environments where adolescents' self-worth becomes contingent on external validation that is both public and unpredictable. Social comparison provides the cognitive process (evaluating self against others), uses and gratifications explains the motivation (seeking connection and validation), and FoMO captures the affective consequence (anxiety about exclusion). Together, these frameworks predict that higher intensity social media use will be associated with increased depressive symptoms, as adolescents become trapped in cycles of comparison, validation-seeking, and exclusion anxiety.

Gender Differences in Adolescent Social Media Use and Mental Health

A critical aspect of social media use among adolescents is the pursuit of validation. The likes, comments, and shares on platforms serve as a form of social reinforcement, contributing to their sense of self-worth. Sherlock and Wagstaff (2019) found that adolescents who heavily rely on this external validation are more prone to emotional distress when their expectations are not met. The absence of positive feedback or the presence of negative interactions can trigger feelings of rejection and exacerbate depressive symptoms. This dependency on external feedback may also lead to compulsive social media use, perpetuating a cycle of vulnerability and negative mental health outcomes (Casale & Fioravanti, 2018). Gender may moderate the gratifications sought from social media and their effects. Females are more likely to use social media for relational purposes, such as maintaining friendships and seeking emotional support, while males often use it for entertainment and information (Muscanell & Guadagno, 2012). These differences can influence how adolescents experience the psychological impacts of social media. For instance, females may be more vulnerable to the negative effects of social comparisons and cyberbullying, while males may be more affected by content related to competition or achievement.

Previous research shows that gender differences exist in both social media engagement and emotional responses. Neuroscientific studies indicate that female adolescents exhibit greater emotional sensitivity to social rejection and peer approval (Silk et al., 2012). Since social media often amplifies social validation processes, female users may experience stronger emotional reactions to both positive and negative feedback, making them more vulnerable to the psychological consequences of online interactions. Females are more likely to engage in ruminative coping strategies, meaning they are more prone to repeatedly focus on negative emotions and experiences (Nolen-Hoeksema, 2012). When exposed to distressing content or negative social comparisons on social media, they may dwell on these experiences longer, exacerbating depressive symptoms.

Gender differences in emotional sensitivity to peer evaluation (neuroscience), ruminative coping (psychology), and relational social media use (behavior) converge to suggest differential vulnerability. We therefore predict that the association between social media use and depressive symptoms will be stronger among females.

Parental Monitoring as a Regulatory Buffer in the Digital Environment

Another important dimension in understanding the effects of social media on adolescent mental health is the role of parental monitoring as a potential moderating factor. Building on Bronfenbrenner's ecological systems theory (1979), parental monitoring can be understood as part of the immediate family environment that shapes how adolescents experience external influences such as social media. Adolescents embedded in regulated family environments may be less susceptible to the adverse psychological consequences of social media use. The parental mediation literature distinguishes between active mediation (e.g., discussions about media use), restrictive mediation (e.g., setting rules), and co-use. Active mediation, in particular, has been shown to mitigate the negative psychological outcomes of online activity by fostering critical thinking and emotional processing (Clark, 2011). Greater digital literacy can enable adolescents to critically evaluate content and avoid maladaptive social comparisons, thereby reducing depressive symptoms.

From a developmental neuroscience perspective, adolescents exhibit heightened sensitivity to peer influence and limited self-regulatory capacity due to ongoing maturation of the prefrontal cortex (Crone & Dahl, 2012). Research has shown that parental involvement can significantly influence adolescents' emotional regulation and behavioral health (Wang et al., 2011). Higher parental monitoring has also been linked to lower engagement in risk behaviors, including problematic technology use (Laird et al., 2003). In contrast, low parental oversight may allow excessive or maladaptive social media use to develop unchecked, increasing vulnerability to depressive outcomes (Riehm et al., 2019; Twenge & Campbell, 2018). Parental awareness of adolescents' online and offline behaviors can serve as a protective factor by reducing exposure to harmful content or interactions, encouraging healthier digital habits.

Integrating ecological systems theory with research on adolescent neurodevelopment and parental mediation, we predict that parental monitoring will buffer the relationship between social media use and depressive symptoms. Higher parental awareness may facilitate early intervention, encourage critical media evaluation, and provide emotional support that reduces dependence on online validation.

Current Work

This study contributes to ongoing research on the mental health consequences of digital media by examining how the intensity of social media use relates to depressive symptoms among adolescents. Building on psychological theories such as social comparison theory, uses and gratifications theory, and ecological systems theory, we focus on mechanisms that link social media engagement to adolescent well-being. Specifically, we investigate whether parental monitoring serves as a moderating factor that may buffer the relationship between social media use and depressive symptoms. In addition, we explore potential gender differences in this association, informed by prior findings on emotional sensitivity and media use motivations among adolescents.

Using nationally representative survey data, our study provides a theory-driven analysis of how digital behaviors and family context interact in shaping adolescent mental health. By addressing both risk and protective factors, this research aims to inform preventive strategies and policy debates concerning adolescent media regulation.

Data and Method

The analysis is based on data from the 2023 US Youth Risk Behavior Survey (YRBS), a nationally representative, school-based survey conducted biennially by the Centers for Disease Control and Prevention (CDC). The YRBS employs a stratified, three-stage cluster sampling design to produce estimates representative of US students in grades 9 through 12 attending public and private schools. In the first stage, primary sampling units (PSUs), comprised of entire counties or groups of smaller adjacent counties, are stratified by region and urbanicity and then selected with probability proportional to school enrollment size. In the second stage, schools within selected PSUs are sampled using probability proportional to enrollment size. In the third stage, classes are randomly selected within chosen schools to ensure representation across grades. Ethical oversight for the YRBS is maintained by the CDC and participating jurisdictions. Data collection protocols include parental consent

procedures, and student participation is voluntary and anonymous. The survey received institutional review board (IRB) approval at the CDC. Additional details on survey methodology and ethical safeguards can be found in Brener (2024). After excluding cases with missing values on variables of interest, the final analytic sample consisted of 10,334 respondents.

The dependent variable captured depressive symptomatology using a single dichotomous item: "During the past 12 months, did you ever feel so sad or hopeless almost every day for two weeks or more in a row that you stopped doing some usual activities?" Responses were coded as 1 ("yes") and 0 ("no"). This item reflects a core diagnostic criterion for major depressive episodes as outlined in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), specifically the experience of persistent low mood and anhedonia leading to functional impairment over a sustained period (American Psychiatric Association, 2013). Although it does not encompass the full symptom profile of clinical depression (e.g., changes in sleep, appetite, or concentration), it serves as a validated and accepted proxy for depressive symptoms in large-scale adolescent health surveys (Brener et al., 2002; Pontes et al., 2020; Verlenden et al., 2024).

The primary independent variable was self-reported frequency of social media use. We collapsed the original eight-category frequency variable (ranging from "I do not use social media" to "more than once an hour") into three groups: low (categories 1–3: non-use to about once a week), moderate (categories 4–5: a few times a week to about once a day), and high (categories 6–8: several times a day or more). Collapsing detailed ordinal measures into a small number of interpretable groups is common in the literature (e.g., Twenge & Campbell, 2018; Foerster et al., 2015). We also report robustness models using the full eight-category scale (Appendix T able A3), where "I do not use social media" is the reference group.

This approach is consistent with prior research that has grouped media use into a small number of categories for analysis, such as Twenge and Campbell (2018), who used four screen-time groups, and Foerster et al. (2015), who classified adolescents into distinct user groups based on patterns of phone and social media use. Responses were grouped into three categories to reflect overall usage patterns: low use (up to once a week), moderate use (a few times a week to once daily), and high use (several times a day or more). This categorization balances interpretability with variation in the data and has been used in prior research to differentiate typical from heavy users.

Parental monitoring was measured with a survey item asking adolescents, "How often do your parents or other adults in your family know where you are going or with whom you will be?" In everyday terms, this reflects how much parents or guardians are aware of their teenagers' plans and social contacts. While the broader literature on parental monitoring distinguishes between different approaches, such as setting limits on screen use, discussing online experiences, or directly checking devices, the YRBS measure specifically captures the awareness dimension of monitoring. We therefore interpret our findings in light of this narrower, but widely used, operationalization. Response options ranged from "never," "rarely," "sometimes," "most of the time," to "always." Responses were reverse-coded so that higher values reflect greater parental oversight.

Several covariates were included to account for potential confounding factors, each of which has been linked to adolescent depressive symptoms in prior research. Gender was coded as a binary variable (0 = female, 1 = male), as numerous studies have documented higher rates of depression among adolescent females (Twenge & Nolen-Hoeksema, 2002). Age was coded in categories ranging from 12 years or younger to 18 years or older, and for analysis, it was treated as continuous. The lowest and highest categories were assigned values of 12 and 18,

respectively; as these corresponded to relatively few, their inclusion does not materially affect results. Electronic bullying was measured with a dichotomous indicator capturing whether the respondent had experienced bullying via digital means, which has been consistently associated with elevated depressive risk (Kowalski et al., 2014). Physical activity was included as a protective behavioral factor, measured by the number of days per week the adolescent was physically active for at least 60 min, given robust evidence that physical activity reduces depressive symptoms (Biddle & Asare, 2011). Finally, school connectedness was assessed using an item capturing students' sense of belonging at school, a factor shown to buffer against depressive outcomes (Resnick et al., 1997). Like parental monitoring, this measure was reverse-coded such that higher scores indicate stronger connectedness.

Binary logistic regression was used to examine the association between social media use and depressive symptoms, adjusting for relevant demographic and psychosocial covariates. A baseline model included main effects for social media use, gender, age, electronic bullying, physical activity, school connectedness, and parental monitoring. To test for moderation, two additional models were estimated. A second model tested gender as a moderator, while a third model assessed the moderating effect of parental monitoring on the relationship between social media use and depressive symptoms. Model estimates are presented as odds ratios (ORs) with 95% confidence intervals. Predicted probabilities were calculated to visualize the interaction effects and aid interpretation. The present analysis did not apply survey weights. As such, findings should be interpreted as descriptive of the unweighted analytic sample rather than as generalizable to the national adolescent population.

To assess potential multicollinearity among predictors, we computed variance inflation factors (VIFs) for all models. In the baseline model, all predictors exhibited VIF values well below 2, indicating no multicollinearity concerns. While VIFs were higher in models involving interaction terms, as is typical (Aiken & West, 1991; Craney & Surles, 2002), generalized VIF values remained below 5, including for the interaction between social media use and parental monitoring. These results suggest that multicollinearity is unlikely to substantially bias the estimated effects.

Little's (1988) MCAR test indicated that the assumption of data missing completely at random was violated ($\chi^2(416) = 1662$, $p < .001$). To investigate the nature of the missingness, we estimated logistic regression models predicting nonresponse for key variables (see Appendix Fig. A1 and Table 1). These analyses revealed that missingness on parental monitoring was systematically associated with social media use and experiences of electronic bullying, while missingness on depressive symptoms was associated with both parental monitoring and social media use. Despite evidence that the data are not MCAR, we retained listwise deletion as our primary analytic approach for two main reasons: (1) the proportion of missing data on the variables of interest was relatively low, and (2) listwise deletion provides unbiased estimates under a missing at random (MAR) mechanism when the missingness is accounted for by the observed variables included in the model (Allison, 2001). Furthermore, we conducted multiple imputation (Van Buuren et al., 2011) as a robustness check, and the results were substantively identical (see Appendix Table 2), confirming that our primary findings are not sensitive to the handling of missing data.

Results

Table 1 presents descriptive statistics for the study sample ($N = 10{,}334$). Approximately 40% of adolescents reported experiencing depressive symptoms in the past year. The majority of respondents (80.3%) reported high levels of social media use, while 10.4% fell into the moderate use category and 9.3% reported low use. Parental monitoring was relatively high overall (M = 4.2, SD = 1.0), while school connectedness had a slightly lower average (M = 3.4, SD = 1.1). About one in five adolescents (19.6%) reported experiencing electronic bullying. On average, participants engaged in physical activity 1.9 days per week (SD = 1.0). The sample was evenly divided by gender, and the average age was just over 16 years (M = 5.1 on a 7-point ordinal scale), corresponding to students between 15 and 17 years of age.

A binary logistic regression model was estimated to examine the association between social media use and depressive symptoms among adolescents, controlling for gender, age, electronic bullying, physical activity, school connectedness, and parental monitoring (see Table 2). For model fit evaluation, we report the Akaike information criterion (AIC), the Bayesian information criterion (BIC), and Tjur's R^2. AIC and BIC are relative measures of model fit, with lower values indicating better fit among the set of candidate models (Burnham & Anderson, 2002). Tjur's (2009) R^2 captures the difference in average predicted probabilities between adolescents with and without depressive symptoms, with higher values reflecting better model discrimination. In our case, differences in AIC and BIC across models were small and did not meaningfully influence model selection; we include these fit indices for completeness alongside the reported R^2 values.

In the baseline model, both moderate and high levels of social media use were significantly associated with increased odds of reporting depressive symptoms compared to low use. Adolescents who reported moderate use had 1.31 times greater odds of depressive symptoms (95% CI [1.07, 1.60], $p < .01$), while those with high use had 1.77 times greater odds (95% CI [1.52, 2.08], $p < .001$).

Several covariates also emerged as significant predictors of depressive symptoms. Male adolescents were significantly less likely than females to report depressive symptoms (OR = 0.36, 95% CI [0.33, 0.40], $p < .001$). Electronic bullying was strongly associated with elevated odds of depressive symptoms, with victims nearly four times more likely

Table 1 Descriptives

Variable	Mean/percentage	SD	Min	Median	Max
Depressive symptoms	0.44	0.50	0.00	0.00	1.00
Parental monitoring	4.22	0.98	1.00	4.00	5.00
School connectedness	3.39	1.14	1.00	4.00	5.00
Physical activity days	1.91	1.05	1.00	2.00	4.00
Age (years)	16.06	1.21	12.00	16.00	18.00
Gender (male)	50.90%				
Electronically bullied	17.30%				
Social media group					
– Low	9.30%				
– Moderate	10.40%				
– High	80.30%				

Table 2 Logistic regression predicting depressive symptoms

Predictor	Baseline	SM × gender	SM × monitoring
(Intercept)	6.75*** [3.50, 10.74]	6.28*** [3.19, 12.37]	4.06*** [1.87, 8.82]
Moderate social media use (vs. low)	1.31** [1.07, 1.60]	1.32 [0.97, 1.80]	3.16** [1.54, 6.54]
High social media use (vs. low)	1.77*** [1.52, 2.08]	1.93*** [1.52, 2.46]	3.10*** [1.84, 5.24]
Male (ref: female)	0.36*** [0.33, 0.39]	0.40*** [0.30, 0.54]	0.35*** [0.32, 0.39]
Age	1.01 [0.98, 1.05]	1.01 [0.98, 1.05]	1.01 [0.98, 1.05]
Electronically bullied (yes)	3.69*** [3.27, 4.16]	3.69*** [3.27, 4.16]	3.69*** [3.27, 4.17]
Physically active (days/week)	0.85*** [0.81, 0.88]	0.85*** [0.81, 0.88]	0.85*** [0.81, 0.88]
School connectedness	0.72*** [0.69, 0.75]	0.72*** [0.69, 0.75]	0.72*** [0.69, 0.75]
Parental monitoring	0.77*** [0.74, 0.80]	0.77*** [0.74, 0.80]	0.87* [0.78, 0.98]
Moderate SM × male	—	0.99 [0.66, 1.49]	—
High SM × male	—	0.86 [0.63, 1.18]	—
Moderate SM × monitoring	—	—	0.80* [0.68, 0.95]
High SM × monitoring	—	—	0.87* [0.77, 0.98]
Num. Obs	10,334	10,334	10,334
AIC	12,187.80	12,190.20	12,184.60
BIC	12,253.00	12,269.90	12,264.30
Tjur's R^2	0.18	0.18	0.18

Note: Odds ratios (OR) shown; 95% confidence intervals in brackets. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

to report symptoms than those who were not bullied (OR = 3.69, 95% CI [3.10, 4.40], $p < .001$). Physical activity showed a protective effect; each additional day of physical activity per week was associated with lower odds of depressive symptoms (OR = 0.85, 95% CI [0.82, 0.88], $p < .001$). Similarly, both school connectedness and parental monitoring were negatively associated with depressive symptoms. Higher school connectedness reduced the odds of depressive symptoms (OR = 0.72, 95% CI [0.70, 0.74], $p < .001$), as did greater parental monitoring (OR = 0.77, 95% CI [0.74, 0.79], $p < .001$). Age was not a significant predictor in the model.

To test whether the association between social media use and depressive symptoms varied by gender, a second model included interaction terms between the two variables (SM × male). Contrary to our hypothesis, these interaction terms were not statistically significant (moderate use × gender: OR = 1.12, 95% CI [0.88, 1.43], $p = .36$; high use × gender: OR = 0.94, 95% CI [0.78, 1.14], $p = .53$). These results suggest that while females report higher overall rates of depressive symptoms, the strength of the association between social media use intensity and depressive symptoms does not significantly differ between male and female adolescents.

Furthermore, a third model included an interaction term for social media use and parental monitoring (SM × monitoring). Here, results showed significant moderation effects: the interaction between moderate social media use and parental monitoring was significant (OR = 0.80, 95% CI [0.68, 0.95], $p < .05$), as was the interaction between high use and parental monitoring (OR = 0.87, 95% CI [0.77, 0.98], $p < .05$). These findings indicate that higher levels of parental monitoring buffer the negative association between social media use and depressive symptoms. As illustrated in Fig. 1, the predicted probability of depressive symptoms increases more sharply with higher social media use among adolescents with low parental monitoring, while this association is attenuated among those with higher

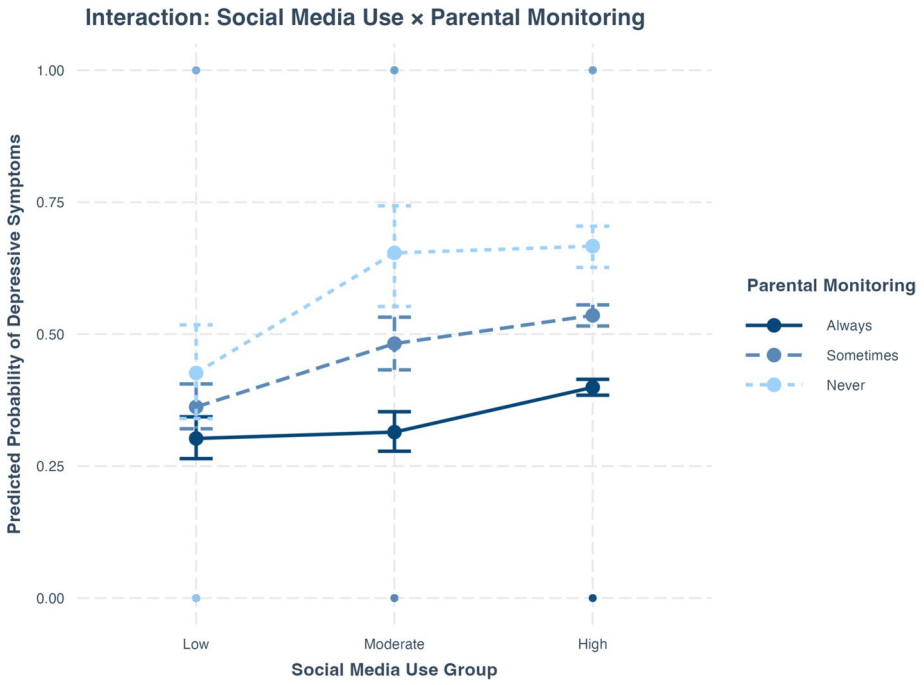


Fig. 1 Visualization of interaction effect

levels of monitoring. Figure 1 displays predicted probabilities of depressive symptoms at low, medium, and high values of parental monitoring, while the underlying regression models were estimated using the full 5-point scale.

To assess the robustness of our findings, we replicated the main regression models using the original eight ordinal response categories for social media use rather than the trichotomized version. As shown in Appendix Table A3 and Fig. A2, the moderating effect of parental monitoring persists under this specification, with particularly pronounced buffering effects observed at medium to high levels of social media engagement. As a robustness check, we furthermore re-estimated the interaction model using a mean-centered version of the parental monitoring variable to reduce potential multicollinearity. The results were consistent with our original findings (see Appendix Table A4). Additionally, we estimated all models using pooled estimates from multiply imputed datasets, which yielded results highly similar to those based on complete case analysis (see Appendix Table 2). These robustness checks support the validity of our findings and demonstrate that the trichotomous grouping used in the main models does not obscure meaningful variation in the data.

Discussion

The results of this study provide evidence consistent with a positive relationship between social media use and depressive symptoms among adolescents, while our second hypothesis about gender moderation was not supported by the data. The findings indicate that both moderate and high levels of social media use are significantly

associated with increased odds of experiencing depressive symptoms, with high-intensity users showing the strongest effects. Furthermore, we found evidence that parental monitoring moderates this relationship, suggesting that appropriate parental involvement may help mitigate the negative consequences of social media engagement.

The observed positive association between social media intensity and depressive symptoms aligns with previous research highlighting the potentially detrimental effects of excessive digital engagement (Primack et al., 2017; Twenge et al., 2018). Several mechanisms may explain the observed association. First, social comparison theory suggests that exposure to idealized representations on social media can trigger feelings of inadequacy. Second, the uses and gratifications framework highlights psychological costs associated with seeking validation online (Sherlock & Wagstaff, 2019). Third, high-intensity social media use may increase anxiety related to fear of missing out (FoMO), disrupting daily functioning (Przybylski et al., 2013). These findings suggest balanced engagement rather than complete abstention might best support adolescent well-being.

Our results revealed significant gender differences in the prevalence of depressive symptoms, with female adolescents having about 2.8 times the odds of reporting such symptoms compared to their male counterparts. This finding is consistent with previous research documenting higher rates of depression among adolescent girls (Nolen-Hoeksema, 2012). However, contrary to our second hypothesis, the interaction between social media use and gender was not statistically significant. This indicates that while females exhibit higher overall vulnerability to depressive symptoms, the incremental risks associated with higher levels of social media use do not appear to be significantly stronger among them compared to boys. This finding challenges some existing assumptions in the literature about gender-specific vulnerabilities to social media effects. Previous research has suggested that female adolescents might be more susceptible to the negative impacts of social media due to differences in usage patterns and coping styles (Muscanell & Guadagno, 2012; Nolen-Hoeksema, 2012).

Perhaps, the most novel contribution of this study is the identification of parental monitoring as a significant moderator of the relationship between social media use and depressive symptoms. As hypothesized, the negative association between social media and our binary indicator of depressive symptoms was attenuated among adolescents who reported higher levels of parental awareness and oversight. This buffering effect was observed for both moderate and high levels of social media use, suggesting that parental involvement can be protective across different usage patterns. Several mechanisms might explain this moderating effect, though we cannot establish causality given our study design. First, parents who are actively involved in their adolescents' lives may be better positioned to recognize signs of problematic social media use or emotional distress, enabling earlier intervention. Second, parental guidance may help adolescents develop healthier digital habits and more critical media literacy skills, reducing vulnerability to negative social comparisons or cyberbullying. Third, strong parent–adolescent relationships may provide emotional support that buffers against the potential negative effects of online interactions (Wang et al., 2011).

Alternative explanations for the observed associations should also be considered. It is possible that unmeasured factors, such as family conflict, peer victimization beyond electronic bullying, or socioeconomic disadvantage, contribute both to adolescents' social media use and their risk of depression. Although we accounted for several important covariates, residual confounding cannot be ruled out.

Conclusion and Future Research

This study provides compelling evidence of a significant association between social media use intensity and depressive symptoms among adolescents, with higher levels of engagement linked to greater psychological risk. However, our findings also highlight the important moderating role of parental monitoring, suggesting that appropriate adult guidance can mitigate the negative impacts of social media exposure.

Additionally, our finding that moderate social media use is less strongly associated with depressive symptoms than high-intensity use suggests that promoting balanced engagement rather than complete prohibition might be a more realistic and effective approach. Educational initiatives that help adolescents develop healthy digital habits and critical media literacy skills may complement regulatory measures in creating safer online environments. For parents, our findings emphasize the importance of maintaining awareness of adolescents' digital activities without necessarily imposing intrusive surveillance. Open communication about online experiences, guidance in developing critical media literacy, and encouraging balanced engagement may be key components of effective parental involvement. Educators can complement these efforts by fostering school environments that promote belonging and connection and incorporating media literacy into curricula.

Beyond these general implications, our findings offer several concrete applications for clinical, school-based, and community-level interventions. In clinical settings, brief screenings that assess not only depressive symptoms but also digital media habits and the degree of parental involvement may enhance early identification of at-risk adolescents. This recommendation aligns with existing clinical guidance from the American Academy of Pediatrics, which encourages routine inquiry into media use during preventive care visits (Council on Communications and Media et al., 2016), and with empirical evidence supporting the integration of social media assessments into adolescent health care (Moreno & Radovic, 2018). In preventive contexts, school counselors and psychologists could develop interventions that teach adolescents emotion regulation strategies specifically tailored to online interactions, such as managing reactions to social feedback or dealing with exclusion from digital peer groups (Verma et al., 2024). Parent-focused psychoeducational programs might emphasize the importance of fostering open communication and modeling healthy digital behavior, rather than relying on restrictive surveillance alone. This recommendation is consistent with research on parental mediation and digital literacy, which shows that active parental involvement enhances adolescents' ability to critically evaluate online content and promotes healthier digital engagement (Livingstone & Helsper, 2008; Nikken & Schols, 2015).

Contrary to our hypothesis, we did not find evidence that the relationship between social media use and depressive symptoms differs significantly by gender. This suggests that prevention efforts should target both genders, even if girls are generally more affected by depression.

Several limitations of this study should be acknowledged. First, the cross-sectional nature of the data precludes causal inferences about the relationship between social media use and depressive symptoms. While our findings are consistent with the hypothesis that excessive social media use contributes to depressive symptoms, it is also possible that adolescents experiencing depression are more likely to turn to social media as a coping mechanism or form of social support (Frison & Eggermont, 2017). Second, our measure of depressive symptoms was a binary indicator rather than a comprehensive clinical assessment. This measure provides limited information about symptom severity or clinical significance. This binary approach may obscure important nuances in the relationship between social media use and mental health, particularly if the effects

are more pronounced for certain types or severities of depressive symptoms. Third, our measure of social media use focused primarily on frequency rather than content or context. Different types of engagement (e.g., passive browsing versus active communication) may have distinct implications for mental health. Specifically, passive consumption of content, such as scrolling through feeds without engaging, has been found to exacerbate feelings of loneliness and depression (Verduyn et al., 2017). Fourth, the study relied on single-item measures for several key constructs, including depressive symptoms and parental monitoring. While such measures are commonly used in large-scale population surveys due to constraints on questionnaire length and respondent burden (Allen et al., 2022), they may lack the psychometric robustness of multi-item scales and offer limited coverage of the construct domain. Finally, while our analysis controlled for several important covariates, unmeasured factors such as personality traits, pre-existing mental health conditions, or family dynamics may influence both social media use and depressive symptoms. For example, neuroticism has been linked to both heavier social media use and increased depressive symptoms (Marino et al., 2018), and family conflict has been identified as a predictor of adolescent mental health problems (Yap et al., 2014). Future research should incorporate these dimensions to more fully capture the complexity of influences on adolescent well-being.

Future research should address these limitations by employing longitudinal designs to establish temporal precedence and potential bidirectional relationships between social media use and depressive symptoms. Additionally, qualitative approaches that explore adolescents' own perspectives on how social media affects their well-being would complement quantitative findings and inform more targeted interventions. Further investigation of the protective mechanisms underlying parental monitoring would also be valuable. Distinguishing between different forms of monitoring (e.g., co-viewing, open communication) could help identify the most effective approaches for supporting healthy social media use. Similarly, exploring how different parenting styles interact with social media engagement would provide more nuanced guidance for families navigating the digital landscape. Future research could also examine potential three-way interactions between social media use, parental monitoring, and gender to assess whether the buffering role of parental involvement differs for male and female adolescents.

As social media continues to evolve and permeate adolescents' lives, ongoing research will be essential for understanding its complex effects on development and mental health. By building on the findings of this study and addressing its limitations, future work can contribute to more effective strategies for supporting adolescent well-being in an increasingly digital world.

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Data Availability The data used in this study are openly available from the US Centers for Disease Control and Prevention (CDC) and can be accessed via the Youth Risk Behavior Surveillance System (YRBSS) website: https://www.cdc.gov/healthyyouth/data/yrbs/index.htm.

Declarations

Ethics Approval This article did not require ethical approval as it involved secondary analysis of publicly available, anonymized survey data. No research involving human or animal subjects was conducted by the authors.

Conflict of Interest The authors declare no competing interests.

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